

PRODUCT & DESIGN SPECIFICATION

ClockWise People

A multi-tenant workforce management platform for time, attendance, leave, overtime, TOIL, approvals, and payroll preparation.

| *"Track Time. Manage People. Prepare Payroll."*

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01 Executive Summary

ClockWise People is a multi-tenant SaaS platform that gives growing companies one secure system of record for employee time, attendance, leave, overtime, TOIL (time off in lieu), approvals, and payroll preparation — replacing the spreadsheets, paper timesheets, and disconnected tools that most small and mid-sized employers currently rely on.

The platform is built around a single idea: workforce time data should be captured once, calculated correctly by rules the company controls, approved through a clear workflow, and locked down before it ever reaches payroll. Every company that signs up gets its own isolated, secure tenant — its own employees, branches, schedules, rules, and reports — while running on the same shared application.

IN ONE SENTENCE

ClockWise People turns scattered attendance records and manual overtime math into a single, auditable, rules-driven workforce system that companies can trust at payroll time.

Who it's for

ClockWise People is aimed at companies with 50–500 employees across one or more branches — particularly operationally complex environments such as multi-branch healthcare practices, clinics, retail groups, and service businesses where staff work shifts, accumulate overtime, bank TOIL, and need leave tracked accurately against South African (and similar) labour requirements.

What makes it different

CAPABILITY	WHY IT MATTERS
Rules-based overtime & TOIL engine	Overtime and TOIL are calculated automatically from configurable company rules instead of manual spreadsheet formulas that drift out of sync between branches.
TOIL as a transaction ledger	TOIL is never a single editable number — it is a full audit trail of every accrual, conversion, and deduction, so balances can always be explained and defended.
Payroll period locking	Once a period is approved, it locks. Nothing slips through after the fact without a logged, reasoned correction.
True multi-tenancy	Every company's data is isolated at the database level, not just hidden in the interface — so the same platform can safely serve many independent organisations.
Branch-aware from day one	Built for organisations that operate across multiple branches or sites with different schedules, not retrofitted onto a single-location model.

02 Problem Definition

Most growing companies — especially multi-branch operations such as clinical practices, retail groups, and service businesses — manage workforce time and payroll preparation through a patchwork of spreadsheets, paper timesheets, instant messages, and manual calculations. This patchwork works at small scale, but breaks down predictably as headcount, branches, and complexity grow.

2.1 The core problem

PROBLEM STATEMENT

Companies cannot reliably answer three basic questions at any given moment: *How many hours did this employee actually work? How much overtime or TOIL do they have? Is this payroll period ready to be paid?* The data needed to answer these questions lives in different places, is calculated inconsistently between branches, and is rarely auditable after the fact.

2.2 Where the pain shows up

PAIN POINT	TYPICAL CAUSE	BUSINESS IMPACT
Inconsistent overtime calculations	Each branch or admin maintains its own spreadsheet formulas, which drift apart over time	Payroll disputes, over- or under-payment, wasted admin hours reconciling differences
TOIL balances nobody trusts	TOIL is tracked as a single running total in a cell, with no history of how it was earned or spent	Employees and managers argue over balances with no record to settle disputes
Late or missing timesheets	No structured submission or reminder workflow	Payroll is delayed or run on incomplete data
No audit trail	Spreadsheet edits overwrite history with no log of who changed what	Cannot defend figures during a CCMA dispute, audit, or employee query
Leave tracked separately from time	Leave forms, sick notes, and attendance logs live in unrelated systems or inboxes	Leave balances are wrong; payroll has to manually cross-check multiple sources
Payroll periods stay "open" indefinitely	No formal locking mechanism after approval	Numbers keep changing after payroll has already run, causing reissued payslips and corrections
Branch managers lack visibility	Reporting is manual and centralised with HR, not delegated	Approval bottlenecks; HR becomes a single point of failure
Compliance risk	No structured record of hours, leave, or overtime against labour law requirements	Exposure during Department of Labour audits or employee disputes

2.3 Why existing tools fall short

Spreadsheets

- No enforced structure or validation
- Formulas break silently when copied or edited
- No approval workflow or audit trail
- Single point of failure — usually one person's laptop
- Does not scale cleanly across branches

Generic HR / payroll software

- Often payroll-first, with time and TOIL as an afterthought
- Rigid overtime rules that don't match local or company-specific policy
- Expensive, complex, built for large enterprises
- Poor multi-branch and approval-delegation support
- Limited transparency into how figures were calculated

2.4 Cost of inaction

Left unaddressed, this problem compounds as a company grows: every new branch multiplies the number of spreadsheets that can drift out of sync; every new employee adds manual reconciliation work; every payroll cycle carries forward unresolved discrepancies. The administrative burden grows faster than headcount, and the risk of a costly payroll or compliance error grows with it.

03 Proposed Solution

ClockWise People replaces fragmented, manual workforce tracking with one secure, rules-driven system that captures time once, calculates correctly, routes approvals to the right people, and locks payroll periods once they are finalised — with a full audit trail behind every figure.

3.1 Solution principles

PRINCIPLE	WHAT IT MEANS IN PRACTICE
Single source of truth	One employee register, one timesheet record, one TOIL ledger per company — not parallel copies across branches
Rules over manual math	Overtime, TOIL, and leave calculations are derived from configurable company rules, applied consistently everywhere
Everything is auditable	Every approval, adjustment, and lock is logged with who, what, when, and the before/after values
Approve, then lock	Payroll periods move through a clear lifecycle and cannot be silently edited once approved
Right access for the right role	Owners, HR admins, branch managers, payroll viewers, and employees each see only what they need
Branch-native design	Branches and departments are first-class concepts, not bolted on later

3.2 How it solves each problem

PROBLEM (FROM SECTION 2)	CLOCKWISE PEOPLE SOLUTION
Inconsistent overtime calculations	Centralised overtime engine applies the same configurable rules company-wide (Section 9)
TOIL balances nobody trusts	TOIL transaction ledger — every hour earned or spent is a logged, attributable entry (Section 7.4)
Late or missing timesheets	Structured submission workflow with statuses, exception reports, and missing-clocking warnings
No audit trail	System-wide audit log capturing every important change (Section 6.7)
Leave tracked separately from time	Leave requests, balances, and time records live in the same employee record and reporting layer
Payroll periods stay open indefinitely	Formal payroll period lifecycle with locking and controlled reopening (Section 6.5)
Branch managers lack visibility	Branch-scoped approval permissions and reporting so managers self-serve within their own branch
Compliance risk	Structured, exportable records of hours, leave, and overtime ready for audit or dispute resolution

3.3 The experience in short

- 1

Capture — Employees clock or submit daily time, request leave, and see their own balances.
- 2

Calculate — The system automatically derives normal hours, overtime, and TOIL from company rules.
- 3

Approve — Branch managers and HR review and approve timesheets, leave, and overtime through a clear workflow.
- 4

Lock — Once a payroll period is reviewed, it locks, protecting the figures payroll will use.
- 5

Report — Clean, exportable reports give HR, managers, payroll, and owners exactly the view they need.

04 Product Overview & Positioning

4.1 Product type

ClockWise People is a **multi-tenant workforce management platform** for companies that need to manage employee time, attendance, leave, overtime, TOIL, approvals, and payroll preparation in one place.

4.2 Core positioning statement

POSITIONING

ClockWise People is a workforce operations platform that helps companies track employee time, manage leave, calculate overtime and TOIL, approve timesheets, and prepare accurate payroll reports.

4.3 Tagline options

- Time. People. Payroll. Aligned.
- Workforce Management, Done ClockWise.
- Track Time. Manage People. Prepare Payroll.
- The Smarter Way to Manage Workforce Time.
- Keep Your Workforce Moving.

4.4 Brand feel

ATTRIBUTE	ATTRIBUTE	ATTRIBUTE
Professional	Reliable	Precise
Calm	Modern	People-focused

05 Users & Roles

ClockWise People is built around five primary roles. Each role has a distinct set of responsibilities and a correspondingly scoped view of the system — nobody sees more than they need to do their job.

Company Owner / Super Admin		Tenant-level control
Responsible for company setup, billing, user management, and system-wide settings.		
Can do	Configure company settings, manage billing, create/remove users, set company-wide rules, view all branches and reports	
Primary goal	Ensure the company's instance of ClockWise People is correctly configured and that the right people have the right access	

HR Admin		Operational control
Manages employees, leave, timesheets, TOIL, overtime, reports, and compliance records.		
Can do	Maintain the employee register, configure work schedules, approve/override timesheets and leave, run reports, manage TOIL adjustments	
Primary goal	Keep workforce data accurate and payroll-ready across all branches	

Branch Manager		Branch-scoped approval
Approves timesheets, leave requests, overtime, and TOIL bookings for assigned branches.		
Can do	Review and approve/reject requests for their own branch(es) only; view branch-level reports	
Primary goal	Keep their branch's approvals current so payroll is never blocked waiting on them	

Payroll Viewer		Read-only, payroll-ready
Can view payroll-ready reports but cannot edit core records.		
Can do	View and export locked payroll period reports and summaries	
Primary goal	Pull accurate, finalised figures into the payroll run without risk of editing source data	

Employee		Self-service
Can submit timesheets, request leave, view balances, view schedules, and track approvals.		
Can do	Clock/submit time, request leave, view own TOIL and leave balances, track status of their own submissions	
Primary goal	Submit accurate time and leave requests with minimal friction, and trust that their balances are correct	

5.1 Role permission matrix

Capability	Owner	HR Admin	Branch Manager	Payroll Viewer	Employee
Company settings	Full	View	—	—	—
Employee register	Full	Full	View (own branch)	—	View (self)
Submit timesheet	—	On behalf of	On behalf of	—	Self
Approve timesheet	Full	Full	Own branch	—	—
Approve leave	Full	Full	Own branch	—	—
TOIL adjustments	Full	Full	—	—	—
Lock payroll period	Full	Full	—	—	—
View reports	All	All	Own branch	Payroll only	Self only
Audit log	Full	Full	—	—	—

06 Core Feature Specification

6.1 Multi-company support

ClockWise People must support multiple companies using the same application, with each company's data fully isolated using `company_id`. Every company-owned record must always belong to exactly one company.

Required company-level settings

Company name	Registration number	Country
Timezone	Payroll cycle	Standard monthly hours
Standard daily hours	Default lunch duration	Overtime rules
TOIL rules	Leave rules	Approval rules

6.2 Employee register

The employee register stores all employee information needed for HR, attendance, leave, and payroll processing.

Employee ID	Full name	Known as
Email	Phone number	Branch
Department	Job title	Employment type
Start date	Employment status	Work schedule
Manager	Hourly rate / monthly salary	Leave profile
Payroll identifier		

6.3 Branches, departments & work schedules

The system allows companies to structure employees by branch and department — for example, branches such as *ePH*, *KG*, *Ermelo*, *Piet Retief*, and departments such as *Admin*, *Clinical*, *HR*, *Finance*. Each employee belongs to exactly one branch and may belong to a department.

Example work schedule

DAY	START	END	LUNCH	PAID HOURS
Monday – Friday	09:00	18:00	1 hour unpaid	8
Saturday	08:00	13:00	—	5

Work schedules must support: fixed schedules, branch schedules, employee-specific schedules, Saturday rules, public holiday handling, part-time schedules, and monthly standard hour caps.

6.4 Timesheets

Employees submit daily time records. Each entry includes date, clock-in time, lunch start/end, clock-out time, branch worked, notes, status, submission date, and approver/approval date.

The system automatically calculates

- Gross hours and lunch duration
- Paid working hours, normal hours, and overtime hours
- Missing clocking, late arrival, and early departure warnings

Timesheet statuses

DRAFT

SUBMITTED

APPROVED

REJECTED

CANCELLED

LOCKED

6.5 Leave management

Employees can request leave across types including annual, sick, family responsibility, maternity, unpaid, TOIL taken, and other company-defined types. Leave is stored and calculated in hours.

Leave request fields

Employee	Leave type	Start / end date
Total hours	Reason	Attachment
Status	Approved by	Approved date

6.6 Overtime engine

ClockWise People calculates overtime based on configurable company rules. Recommended defaults: an 8-hour standard day, a configurable standard month (e.g. 173.33 or 200 hours), overtime starting only once the normal hour requirement is met, unpaid lunch, and customisable Saturday and public holiday rules.

The overtime engine calculates

Daily paid hours	Monthly paid hours	Normal hours
Excess hours	Overtime before cap	Overtime after cap
Fill-up applied	Payroll overtime	TOIL-convertible overtime

6.7 TOIL engine

TOIL works as a **transaction ledger**, not a single editable balance field — every accrual, conversion, deduction, or correction is a discrete, attributable record.

Example TOIL transactions

TRANSACTION	HOURS
Opening balance	+12
Overtime converted to TOIL	+4
TOIL taken	−8
Manual adjustment	+2
Correction	−1

Each transaction stores: employee, company, date, transaction type, hours, source record, reason, created by, and approved by. The TOIL balance is always *calculated* from the sum of transactions, never stored as an independently editable number.

6.8 Approval workflow

The system supports approval workflows for timesheets, leave requests, overtime, TOIL bookings, and manual adjustments — all sharing the same underlying status model:

DRAFT

SUBMITTED

APPROVED

REJECTED

CANCELLED

LOCKED

Each approval record tracks: request type, request ID, submitted by, approver, approval status, notes, date submitted, and date actioned.

6.9 Payroll period locking

Once a payroll period has been reviewed and approved, it is locked, preventing normal edits. After locking, only an explicit adjustment request, an admin correction with a recorded reason, or an audit log entry can touch the data.

Payroll period lifecycle

- 1

Open — period is active and accepting timesheet/leave activity
- 2

In Review — HR/branch managers reconcile and approve outstanding items
- 3

Approved — figures are confirmed correct
- 4

Locked — no further edits without a logged correction
- 5

Reopened (exception path) — explicitly reopened with reason, fully audited

6.10 Reports

ClockWise People generates clean reports for HR, branch managers, payroll, and company owners, each exportable to PDF, Excel, and CSV.

Monthly payroll summary	Employee timesheet report	Leave balance report
Overtime accumulation report	TOIL balance report	TOIL bookings report
Branch attendance report	Exceptions report	Missing clockings report
Late submissions report	Approval pending report	

6.11 Audit trail

Every important action is logged.

Every important action is logged. Audit logs capture user, company, action, affected table, record ID, old value, new value, timestamp, and IP/device metadata where possible — for example: a timesheet edit, a leave approval, a TOIL adjustment, a payroll period lock, or an employee record update.

07 Key User Flows

7.1 Employee: daily time capture

- 1 Employee clocks in at shift start (or opens a draft timesheet entry)
- 2 Employee records lunch start and lunch end
- 3 Employee clocks out at shift end
- 4 System calculates gross hours, paid hours, and flags any warnings (late arrival, missing clock-out, etc.)
- 5 Employee submits the timesheet for approval — status moves from **DRAFT** to **SUBMITTED**

7.2 Branch manager: timesheet approval

- 1 Branch manager opens the pending approvals queue, scoped to their branch
- 2 Manager reviews flagged exceptions (missing clockings, late arrivals, unusual overtime)
- 3 Manager approves or rejects each timesheet, optionally with notes
- 4 Approved timesheets move to **APPROVED** and feed into the monthly summary; rejected ones return to the employee as **REJECTED**

7.3 Employee: leave request

- 1 Employee selects leave type, date range, and reason; attaches supporting documentation if required
- 2 System calculates total hours requested against the employee's work schedule
- 3 System checks the request against the employee's current leave balance
- 4 Request routes to the branch manager or HR admin for approval
- 5 On approval, the leave balance updates and the days appear on the relevant attendance and payroll reports

7.4 HR admin: month-end payroll preparation

- 1 HR admin opens the current payroll period, status **IN REVIEW**
- 2 HR admin works through the exceptions, missing clockings, and pending approvals reports until the queue is clear
- 3 HR admin reviews the monthly summary per employee — normal hours, overtime, TOIL movement, leave taken
- 4 HR admin marks the period **APPROVED**
- 5 HR admin (or owner) locks the period — status **LOCKED** — generating the final payroll export
- 6 Payroll viewer exports the payroll-ready report for processing in the company's payroll system

7.5 HR admin: TOIL adjustment

- 1 HR admin opens an employee's TOIL ledger
- 2 HR admin creates a manual adjustment transaction with a required reason

3 Adjustment is logged with created-by and (if a second approver is configured) routed for approval

4 Adjustment appears in the employee's TOIL transaction history and recalculates their running balance

5 Action is recorded in the audit trail

08 Data Model & Architecture

8.1 Core tables

companies	company_settings	branches
departments	users	roles
permissions	user_roles	employees
work_schedules	schedule_days	timesheets
time_entries	leave_types	leave_requests
leave_balances	overtime_records	toil_transactions
approval_requests	payroll_periods	monthly_summaries
audit_logs		

8.2 The multi-tenancy rule

NON-NEGOTIABLE RULE

Every company-owned table must include a `company_id` column, enforced at the database layer through Row Level Security — never assumed or enforced only in application code.

```
company_id uuid not null
```

8.3 Entity relationships (conceptual)

At a high level, the data model nests as follows: a **company** owns many **branches**, which contain **departments** and **employees**. Each **employee** has one active **work_schedule**, submits many **timesheets** (each composed of daily **time_entries**), requests **leave_requests** against a **leave_balances** record per leave type, and accrues/spends hours through **toil_transactions**. Approvals for any of these flow through a shared **approval_requests** table. Time and leave data roll up into **monthly_summaries** scoped to a **payroll_period**, and every meaningful change anywhere in the system writes an entry to **audit_logs**.

8.4 Recommended indexes

```
create index idx_employees_company_id on employees(company_id);

create index idx_time_entries_company_employee_date
on time_entries(company_id, employee_id, work_date);

create index idx_leave_requests_company_status
on leave_requests(company_id, status);

create index idx_toil_transactions_company_employee_date
on toil_transactions(company_id, employee_id, transaction_date);

create index idx_monthly_summaries_company_period
on monthly_summaries(company_id, payroll_period_id);
```

09 Business Logic Engine

The calculation engine is the most important part of the system. It must be implemented as a set of pure, independently testable functions — never relied upon only as UI-layer arithmetic — so that every figure on a payslip can be traced back to a deterministic, unit-tested calculation.

```
calculateWorkedHours()
calculateLunchDuration()
calculateDailyNormalHours()
calculateDailyOvertime()
calculateMonthlyNormalHours()
calculateMonthlyOvertime()
calculateToilEarned()
calculateToilTaken()
calculateToilBalance()
calculateLeaveDeduction()
calculatePayrollSummary()
lockPayrollPeriod()
```

9.1 Design requirements for these functions

- **Pure:** given the same inputs, always return the same outputs — no hidden state, no side effects.
- **Testable in isolation:** each function should be unit-testable without touching the database or UI.
- **Rule-driven:** functions read company-level configuration (standard hours, overtime rules, TOIL rules) rather than hard-coding assumptions.
- **Composable:** daily calculations feed monthly calculations, which feed payroll summaries — each layer built on the one below it.
- **Single source of truth:** the UI displays results from these functions; it never recalculates independently.

10 Security & Compliance

ClockWise People must be secure by default, not secured as an afterthought. Because the platform holds sensitive HR and payroll data across multiple independent companies, security controls are treated as core architecture, not optional configuration.

10.1 Required security controls

CONTROL	PURPOSE
Row Level Security on all company-owned tables	Database-enforced tenant isolation — no query can return another company's data, even by mistake
Role-based access control	Each role (Section 5) sees only the data and actions appropriate to it
Company-level data isolation	No record can exist without an owning <code>company_id</code>
Secure authentication	Industry-standard auth with session management
Server-side validation	All input is validated server-side, never trusting client-side checks alone
Zod schema validation	Strongly typed validation for every form and API boundary
Audit logs	Full traceability of sensitive changes (Section 6.11)
No direct cross-company access	Enforced structurally, not just through UI restrictions
Protected admin & API routes	Authorization checks on every privileged route
Payroll-period locking	Prevents silent post-approval edits (Section 6.9)
Soft delete for sensitive records	Preserves history for audit and compliance rather than hard-deleting HR/payroll data

10.2 Why this matters for this category of product

Workforce and payroll data is among the most sensitive data a company holds — it includes personal information, remuneration, and records that may be scrutinised in a labour dispute or compliance audit. A breach, a cross-tenant data leak, or an unauditable change to payroll figures would be a severe trust failure for a multi-tenant platform. Security and auditability are therefore treated as MVP-level requirements, not future enhancements.

11 Performance & Scalability

ClockWise People should comfortably support 50 to 500 employees per company without performance degradation, while remaining efficient as the number of tenant companies on the platform grows.

11.1 Performance practices

- Use proper database indexes (see Section 8.4) on every high-traffic query path
- Filter by `company_id` in all major queries to keep query plans tenant-scoped and efficient
- Paginate large tables (employee lists, timesheet history, audit logs)
- Maintain monthly summary tables rather than recalculating full payroll live on every page load
- Use background jobs or scheduled summary generation for heavy aggregation work
- Use connection pooling in production
- Cache dashboard-level data where appropriate
- Keep large exports server-side rather than building them in the browser
- Avoid loading all employees or timesheets into memory at once

12 Technology Stack

12.1 Frontend

Next.js App Router	TypeScript	React
SCSS Modules or Tailwind CSS	BEM naming (if SCSS)	Framer Motion (subtle motion only)
React Hook Form	Zod validation	TanStack Table
Recharts (dashboards)		

12.2 Backend

Supabase	PostgreSQL	Supabase Auth
Supabase Row Level Security	Supabase Storage	Supabase Edge Functions
SQL migrations		

12.3 Local development

Docker	Supabase CLI	Local Supabase stack
Local PostgreSQL (via Supabase)	GitHub for version control	

12.4 Testing

Vitest	Business logic unit tests
React Testing Library	UI component tests
Playwright	End-to-end testing

12.5 Deployment

Vercel	Next.js frontend hosting
Supabase Cloud	Database, auth, storage
GitHub Actions	CI checks

13 Design System & UI Direction

ClockWise People should feel clean, calm, and professional — an interface that earns trust through clarity rather than decoration, given how much weight its numbers carry at payroll time.

13.1 Design direction

- Minimal SaaS interface, dashboard-first layout
- Clear, consistent status badges across timesheets, leave, and approvals
- Soft shadows and rounded cards
- Spacious, readable tables — never cramped, given how data-dense this product is
- Mobile-first employee screens (clocking in/out, leave requests happen on the move)
- Desktop-first admin screens (approvals, reports, configuration happen at a desk)
- Clear, unambiguous approval states throughout
- No unnecessary animation — motion should clarify state changes, not decorate them

13.2 Suggested colour palette

COLOUR	USE	SWATCH
Deep navy	Trust, primary brand colour, headers	#0B1F3A
Teal	Action, freshness, primary buttons and links	#1A7F73
Soft green	Approved states	#1E8E6F
Amber	Pending states	#D99B2B
Red	Rejected / error states	#C2552F
Light grey	Dashboard backgrounds	#F1F3F5

13.3 Status badge language

A single, consistent badge vocabulary is used everywhere a record has a lifecycle — timesheets, leave, overtime, TOIL bookings, and payroll periods all share the same visual language so users only have to learn it once:

DRAFT

SUBMITTED

APPROVED

REJECTED

CANCELLED

LOCKED

14 Application Structure

A predictable, conventional Next.js App Router structure keeps the codebase navigable as the product grows, and separates business logic cleanly from presentation.

```
/app
  /(auth)
    /login
    /register
  /(dashboard)
    /dashboard
    /employees
    /branches
    /timesheets
    /leave
    /overtime
    /toil
    /payroll
    /reports
    /settings

/components
  /ui
  /forms
  /tables
  /layout
  /charts

/lib
  /supabase
  /auth
  /validations
  /calculations
  /permissions
  /reports

/styles
  /abstracts
  /components
  /layouts
  /pages

/supabase
  /migrations
  /seed.sql
```

Note in particular that `/lib/calculations` is kept separate from `/components` — this is a deliberate enforcement of the principle in Section 9 that business logic must never live only inside UI components.

15 MVP Scope & Roadmap

15.1 MVP scope

The first usable version of ClockWise People should include everything required to run one full payroll cycle end to end, for one company, across multiple branches:

✓ Login	✓ Company setup	✓ Employee register
✓ Branch setup	✓ Work schedules	✓ Timesheet capture
✓ Timesheet approval	✓ Monthly summary	✓ Overtime calculation
✓ TOIL tracking	✓ Leave requests	✓ Payroll export
✓ Audit logs		

15.2 Future features (post-MVP)

Shift planning	Employee documents	Recruitment
Digital onboarding	Induction tracking	Performance reviews
Asset tracking	Policy acknowledgements	Training records
Mobile app	WhatsApp notifications	Payroll software integrations
Biometric clocking integrations	Employee self-service portal	

15.3 Suggested phasing

PHASE	FOCUS	OUTCOME
Phase 1 — Foundation	Auth, multi-tenancy, company/branch setup, employee register	A company can be onboarded and staffed in the system
Phase 2 — Time & Leave	Work schedules, timesheets, leave requests, approval workflow	Day-to-day attendance and leave can be captured and approved
Phase 3 — Overtime & TOIL	Overtime engine, TOIL ledger, monthly summaries	Accurate overtime and TOIL figures are available for every employee
Phase 4 — Payroll & Reporting	Payroll period locking, exports, full report suite, audit log	A full payroll cycle can run end to end with confidence
Phase 5 — Expansion	Post-MVP features per Section 15.2, prioritised by customer demand	Platform grows beyond core time/payroll into broader HR operations

16 Success Metrics

Because ClockWise People exists to replace error-prone manual processes, success should be measured primarily in accuracy, time saved, and trust — not just usage volume.

Metric	What it tells us
Time-to-payroll-lock per period	Whether the approval workflow is reducing the admin burden of closing a payroll period
Number of post-lock corrections	Whether figures are right the first time, or still need frequent fixing after approval
% of timesheets submitted on time	Whether employees are adopting the self-service capture flow
% of approvals completed within SLA	Whether branch managers are keeping pace with their approval queue
TOIL/overtime dispute rate	Whether the transaction ledger is succeeding in making balances trustworthy
Report export usage	Whether HR/payroll are relying on the platform as their source of truth, rather than re-building reports elsewhere
Audit log query frequency	Whether the audit trail is actually being used to resolve disputes or support audits

17 Risks & Open Questions

17.1 Key risks

RISK	MITIGATION
Overtime/TOIL rules vary significantly by company or jurisdiction	Keep the rules engine fully configurable per company rather than hard-coded; validate against South African labour law requirements as a baseline
Row Level Security misconfiguration leaks data across tenants	Mandatory RLS policy tests in CI for every company-owned table before merge
Employees resist moving off familiar spreadsheets	Prioritise a low-friction mobile capture experience and clear, immediate balance visibility
Payroll period locking is too rigid for legitimate late corrections	Provide a controlled, audited reopening path rather than disabling locking entirely
Performance degrades as report complexity grows	Lean on monthly summary tables and background jobs rather than live recalculation (Section 11)

17.2 Open questions for stakeholders

- Which payroll systems (if any) does ClockWise People need to integrate with directly, versus exporting CSV/Excel for manual import?
- Should overtime and TOIL rule templates be pre-built for South African labour law, with other jurisdictions added later, or should the rules engine be jurisdiction-agnostic from day one?
- What is the expected approval SLA per role, and should the system send automated reminders/escalations when it is breached?
- Should biometric clocking integration be considered for an earlier phase given the multi-branch, in-person nature of target customers?
- What level of customisation should company owners have over the approval workflow itself (e.g. single vs. dual approval)?

18 Development Principles

These principles should guide every architectural and product decision throughout the build:

- ✓ **Multi-tenant from day one** — never retrofit isolation onto a single-tenant assumption
- ✓ **Secure by default** — RLS, RBAC, and validation are not optional add-ons
- ✓ **Mobile-friendly for employees** — clocking and leave requests happen on the move
- ✓ **Powerful on desktop for admins** — approvals, reports, and configuration need a full screen
- ✓ **Business logic separated from UI** — calculations live in `/lib/calculations` , not components
- ✓ **Every calculation testable** — pure functions, unit-tested independently of the UI
- ✓ **Every important change auditable** — nothing sensitive changes without a log entry
- ✓ **Every payroll period lockable** — finalised figures are protected from silent edits
- ✓ **Every report exportable** — PDF, Excel, and CSV as standard
- ✓ **Every company isolated** — no cross-tenant access, ever, under any circumstance

19 Final Product Statement

ClockWise People is a modern workforce management platform that helps companies manage employee time, attendance, leave, overtime, TOIL, approvals, and payroll preparation from one secure system.

It is designed for growing companies that need better control, cleaner reporting, and reliable workforce data — without relying on scattered spreadsheets or disconnected tools.